

Q. D. Choose the right answers and label them in your answer sheet.

Final Term Exam - Model 2

Course Name: Software Reengineering
Level : Software Engineering
Time Allowed: 2 Hours

1. **Software aging in reengineering refers to the phenomenon where:** (40x 1 Marks)
☒ a. Software becomes obsolete due to new hardware.
☒ b. Software performance degrades and failure rate increases over time.
c. The developers of the software retire.
d. The software' source code is lost.
2. **Which process is best described as "cleaning up code" to make it easier to understand and maintain without adding new functionality?**
a. Reengineering b. Reverse Engineering ☒ c. Refactoring d. Forward Engineering
3. **When performing refactoring, what must remain unchanged?**
a. The internal code structure.
b. The database schema.
☒ c. The external behavior/functionality.
d. The number of comments.
4. _____ : The level of abstraction of the representation of the system is gradually decreased by successively replacing some aspects of the system with more details.
a. abstraction ☒ b. refinement c. All of them d. None of them
5. _____ : Follows a sequence of activities: formulating concepts about the system to identifying requirements to designing the system to implementing the design.
a. Forward engineering b. reverse engineering c. All of them d. None of them
6. **According to Lehman's Law of Continuing Change, what happens if a software system is not maintained?**
a. It becomes more stable over time.
☒ b. It becomes progressively less useful.
c. It automatically refactors its own code.
d. Its complexity decreases.

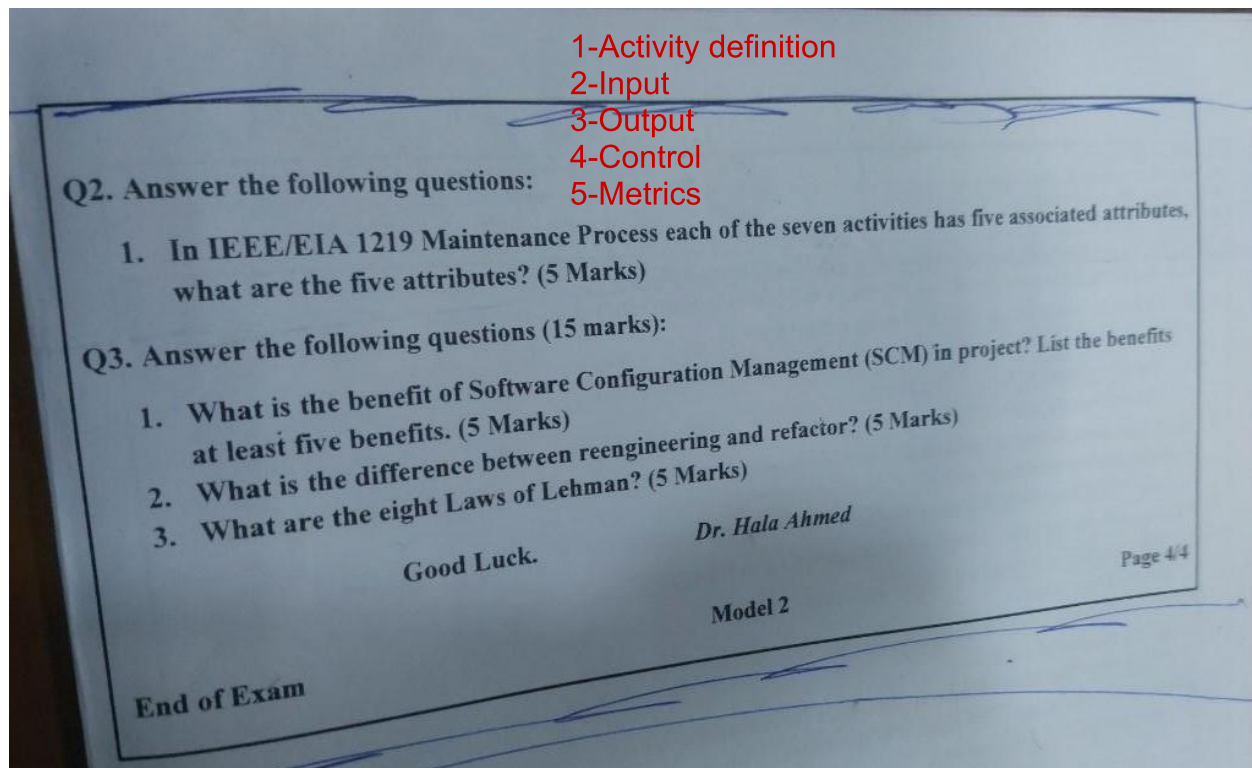
7. **When does one decides to re-engineer a product?**
a. when tools to support restructuring are disabled b. when system crashes frequently
☒ c. when hardware or software support becomes obsolete d. subsystems of a larger system require few maintenance
8. **Every developer maintains at least one workspace. Therefore, the developer makes modifications to the source code, compiles the files, performs tests, and debugs code without impacting the works of other developers.**
a. Sandbox b. Building ☒ c. Isolation d. None of them
9. **Components refer to the ability to enhance the functionality of a component is known as.....**
a. Wrapper b. Glue ☒ c. Tailoring d. None of them
10. **Maintenance is performed on an operational system. Therefore, all modifications must be compatible with the constraints of the existing architecture, design, and code.**
a. Constraints of an existing system b. shorter time frame c. Available test data d. None of them
11. **In software reengineering the word "product items" include**
a. Document b. executable software c. Source code, hardware and disks. ☒ d. All of them
12. **Which of the following is not a business goal of reengineering?**
a. Cost reduction ☒ b. Time reduction ☒ c. Maintainability d. None of the mentioned

13. _____ is an old system which is valuable for the company which often developed and owns it.
 a. Re-engineering analysis ~~b. A legacy system~~ c. Software Configuration Management d. Impact
14. _____ strategy reflects the principle of alteration. By means of alteration, an operational system is transformed into a new system, while preserving the abstraction level of the original system
 a. Rewrite ~~b. rework~~ c. replace. d. None of them
15. The system is no longer a key asset for the developers, who concentrate mainly on maintenance tasks rather than architectural or functional change. Knowledge about the system has lessened and the effects of change have become harder to predict. The costs and risks of change have increased significantly.
 a. Phase out ~~b. Servicing~~ c. Evolution d. Initial development
16. The amount of new content in each successive release of a system tends to stay constant or decrease over time.
 a. Declining quality b. Continuing growth ~~c. Conservation of familiarity~~ d. Conservation of organizational stability.

17. _____ is the process of making a change to the internal structure of software to make it easier to understand and cheaper to modify without changing its observable behavior.
~~a. Refactoring~~ b. E-type c. P-type d. SCM
18. Checked out files are put in a workspace to be freely edited. In addition, it is not necessary to lock the original files in the repository.
 a. Sandbox b. Building c. Isolation d. None of them
19. Software maintenance has some unique characteristics:
 a. Constraints of an existing system b. Shorter time frame c. Available test data ~~d. All of them~~
20. It has been decided to replace or eliminate the system entirely, either because the costs of maintaining the old system have become prohibitive or because there is a newer solution waiting in the pipeline.
 a. Phase out b. Servicing c. Evolution d. Initial development.
21. _____ key concepts used in software development, and both the concepts are equally useful in reengineering.
 a. Abstraction only b. Refinement only ~~c. Abstraction and Refinement~~ d. None of them

22. Software systems are reengineered by keeping one or more of the following general objectives in mind:
 a. Migrating to a new technology. b. Improving maintainability.
 c. Preparing for functional enhancement. ~~d. All of them.~~
23. A system will become progressively less satisfying to its user over time, unless it is continually adapted to meet new needs.
 a. Continuing change b. Self-regulation c. Conservation of organizational stability d. Increasing complexity.
24. _____ means that there are many modules or components in a system which are not used in the delivery of the business functions of the system.
 a. Pollution b. Poor lexicon c. Embedded knowledge d. Coupling
25. It is a piece of code that one builds to isolate the underlying components from other components of the system.
 a. Wrapper b. Glue c. Tailoring d. None of them
26. There are types of aging in software lifecycles which is correct.
 a. software process execution aging b. software product aging ~~c. All of them~~ d. None of them

27. The component provides the functionality to combine different components is known as
 a. Wrapper ☒ b. Glue c. Tailoring d. None of them
28. The code is said to have decayed if it is very difficult to change it, as reflected by many key responses:
 a. the cost of the change b. the calendar or clock time to make the changes c. the quality of the changed software ☒ d. All of Them
29. In this model, a new system is built from components of the old system and others available in the repository.
 a. Quick fix model b. Iterative enhancement model ☒ c. Full reuse model d. All of them
30. This refers to the items that are measured during the execution of the activity.
 a. Control ☒ b. Metrics c. Activity definition d. None of them
31. _____ is important when an application is shifted from one platform to another. So, portability, reusability and adaptability come into this area.
 a. Operational ☒ b. Transitional c. Maintenance d. None of them
32. _____ means a continuous change from lesser, simpler, or worse state to a higher or better state.
 a. Software maintenance ☒ b. Software evolution c. Rewrite d. None of them
33. _____ An organization may decide that supporting software is not core business and may outsource it to a specialist organization offering this service.
 a. Freeze ☒ b. Outsource c. Carry on maintenance ☒ d. All of them
34. _____ is the task of estimating the parts of the software that can be affected if a proposed change request is made.
 a. Re-engineering b. A legacy system c. Software Configuration Management ☒ d. Impact analysis
35. _____ in software development, test cases are designed from scratch, whereas software maintenance can select a subset of these test cases and execute them as regression tests.
 a. Constraints of an existing system b. Shorter time frame ☒ c. Available test data d. None of them
36. _____ In this model, the first changes are made to the highest level documents. Eventually, changes are propagated down to the code level.
 a. ☒ Iterative enhancement model b. Quick fix model ☒ c. Full reuse model d. None of them
37. An environment that enables the maintainer to make and test the changes in an isolated manner is called _____
☒ a. Workspace control b. Version control c. Change management d. None of them
38. _____ is an example of Reverse Engineering, in which object code is translated into a high-level program.
☒ a. Decompilation b. Quick fix model c. Iterative enhancement model d. None of them
39. _____ means reengineering of source code without addition or reduction of features in the transformed targeted source code.
☒ a. Rehosting b. Workspace control c. Change management d. None of them
40. _____ a system means manipulating the concepts embodied in an existing system to create a system that operates in a different problem domain.
 a. Recode ☒ b. Re-thinking c. Re-specify ☒ d. Redesign



q3-->1.

- 1-Life cycle cost is reduced.
- 2- Productivity is improved,
- 3-Quality is ensured
- 4-Correct product configurations are ensured
- 5-Confusion is reduced

q3-->2.

refactor: is the process of making a change to the internal structure of software to make it easier to understand and cheaper..

Re-engineering is the examination and alteration of a system to reconstitute it in a new form...

q3-->3.

- 1-Continuing change
- 2-Increasing complexity
- 3- Self-regulation
- 4-Conservation of organizational stability
- 5- Conservation of familiarity
- 6- Continuing growth
- 7-Declining quality
- 8-Feedback system